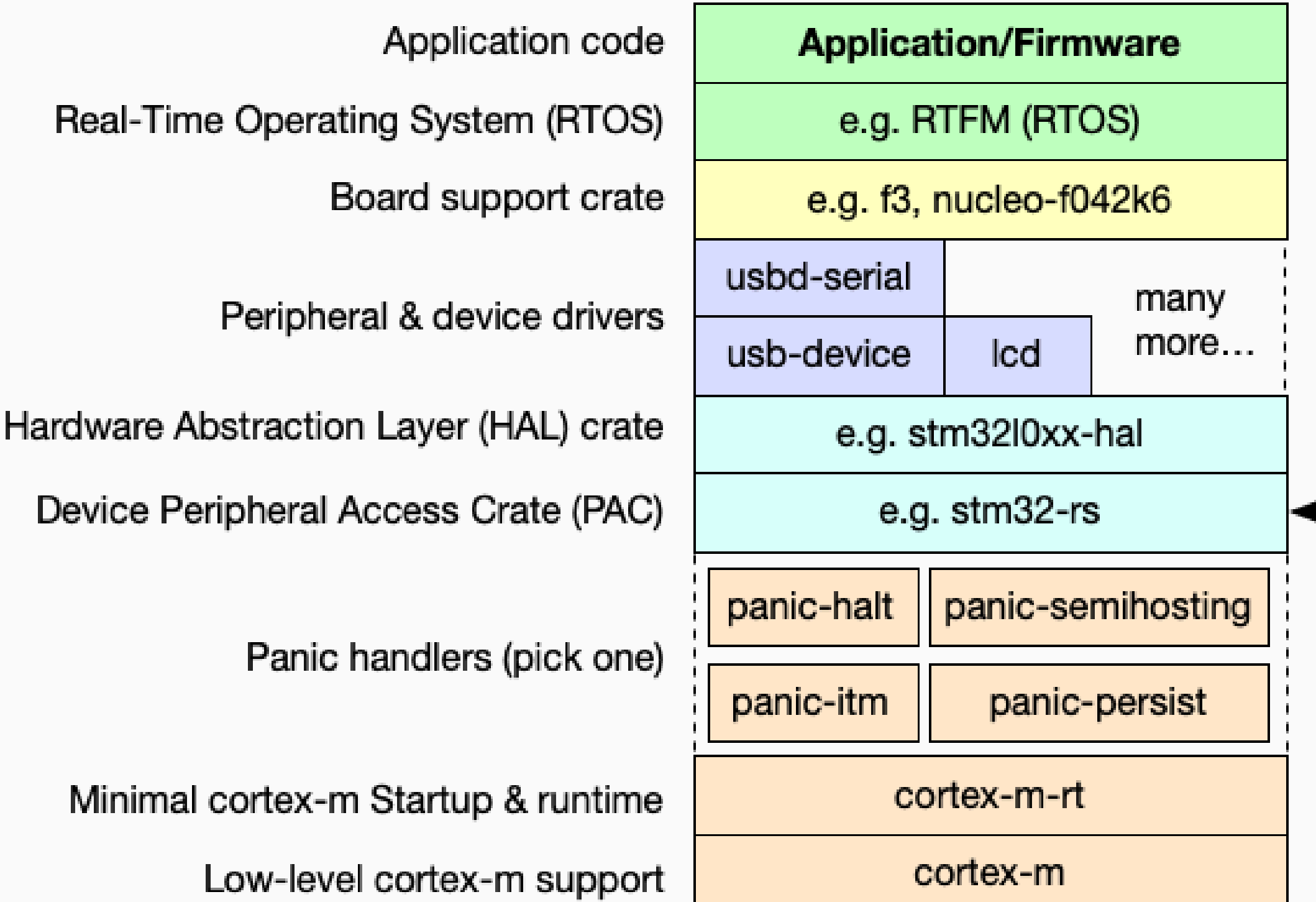
Rust pour le déve
d'applications mé
haut-niveau ! 🦀

Rust 🦀 est connu pour être un langage qui favoriserait le "bas-niveau"...



BDX I/O

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Coercing Arc<Mutex<Option
Arc<Mutex<Option<Box<dy

Asked 4 years, 4 months ago Modified 4 years, 4 months a

3

I'm trying to store a dyn trait inside `Arc<Mute` reason it won't work

```rust
use std::sync::{Arc, Mutex};

trait A{}

struct B{}

impl A for B{}
```

HEY, J'SUIS PAS VENUE
SOUFFRIR, OK ??

I am once again asking

🦀Rust🦀
pour des applications métiers

BDX I/O

Un système de types expressif

La base: Structs, et "Newtypes"

Un cas ultra-classique:

```rust
pub struct User {
    pub email: String,
}

fn ma_fonction(user: User) {
  if !user.email.contains("@") {
     // Gestion de l'erreur, etc.
  }
  // On continue...

  // Possible, mais pas voulu -> Bug !
  let city: String = user.email
}
```

Encoder la logique métier, le rêve ! 😌

```rust
use crate::domain::UserEmail;

pub struct User {
    // Attendez…c'est quoi ça ?
    pub email: UserEmail,
}
```

On "impl"émente la logique pour traiter les cas :

```rust
use validator::ValidateEmail;

#[derive(Debug)]
pub struct UserEmail(String);

impl UserEmail {
    pub fn parse(s: String) -> Result<UserEmail, String> {
        if s.validate_email() {
            Ok(Self(s))
        } else {
            Err(format!("{} is not a valid user email", s))
        }
    }
}
```

Comment on valide ça ? Avec des tests ! 🤩

```rust
#[cfg(test)]
mod tests {
    #[test]
    fn email_missing_at_symbol_is_rejected() {
        let email = "stephanedomain.com".to_string();
        assert_err!(UserEmail::parse(email));
    }

    #[test]
    fn email_missing_subject_is_rejected() {
        let email = "@domain.com".to_string();
        assert_err!(UserEmail::parse(email));
    }
}
```

Un newtype n'est pas égal à un autre newtype !

```rust
pub struct Address(String);
pub struct UserEmail(String);

pub struct User { pub email: UserEmail }

let my_user = User {
  // 💥Erreur ! Pas un UserEmail, mais un String !
  email: "steph@mydomain.com"
}

let my_user = User {
  // 💥Erreur ! Pas un UserEmail, mais un Address !
  email: Address("steph@mydomain.com")
}
```

Il faut utiliser le newtype prévu :

```rust
use crate::domain::UserEmail;

pub struct User {
    pub email: UserEmail,
}

let my_user = User {
  // Ok 👍
  email: UserEmail::parse("steph@mydomain.com")?;
}
// Note: Pour l'instant le `?` c'est "magique" 🪄
```

Parse, don't validate

2019-11-05 ◉ functional programming, haskell, types

Historically, I've struggled to find a concise, simple way to ex
type-driven design. Too often, when someone asks me "Hov
approach?" I find I can't give them a satisfying answer. I kno
vision—I have an iterative design process that doesn't requir
of thin air—yet I haven't been very successful in communica

However, about a month ago, I was reflecting on Twitter abo
parsing JSON in statically- and dynamically-typed language
looking for. Now I have a single, snappy slogan that encapsu
means to me, and better yet, it's only three words long:

Newtype Patterns (Experience I

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Abstract

We investigate the effects on code quality of the design patterns typestate and newtype in the language Rust, by conducting three case studies from production code, evaluated through expert interviews, static code analysis and benchmarks. The findings indicate that: (1) Typestate improves code faultlessness and testability, but comes at the cost of more boilerplate code and can degrade readability. Code with extensive branching logic and many invariants likely benefits most from the pattern. (2) Newtype combined with the "Parse, don't validate" principle improves code quality at a low cost and prevents invalid states at runtime.

CCS Concepts: • **Software and its engineering** → *Patterns.*

Keywords: Rust, design patterns, typestate, newtype, func-

Most catalogued desig
OOP, for example the Gar
GoF patterns become ob
transferred to Rust, since
native constructs for the

Rust-specific patterns s
emerged. The Rust open
few of them in rust-unof
four recurring functional
strates their benefits thr
These resources provide
evidence on the patterns'

backend applications, guided by a method built on SQuaRE criteria and combining execution-time benchmarks, static code analysis and expert interviews.

Neither typestate nor newtype incurred additional runtime cost. Typestate clearly improved testability and faultlessness by shifting error detection to compile time. However, its effect on readability is contested and is not captured by the static metrics. The additional boilerplate is perceived as cognitive overhead by some experts, while others find the explicit states easier to follow. Combined with the principle "Parse, don't validate", the newtype pattern introduced minimal downsides and code changes while improving almost all quality dimensions, making it broadly applicable.

Beyond addressing the limitations from subsection 4.4, several research directions open up. Because typestate's main downside is boilerplate, future studies should investigate macro rules or libraries that generate the state scaffolding and let developers focus on the logic of each state transition,

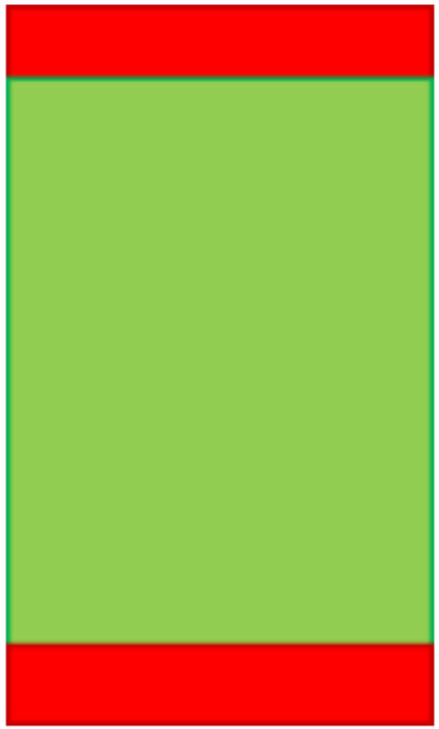
Since January 2017 I've been singing the praise of the *impure/pure/impure* sandwich, but I've never published an article that defines the term. I intend this article to remedy the situation.

Functional architecture

In a functional architecture, pure functions can't call impure actions. On the other hand, as Simon Peyton Jones observe in a lecture, *observing the result of pure computation is a side-effect*. In practical terms, *executing* a pure function is als impure, because it happens non-deterministically. Thus, even for a piece of software written in a functional style, the entry point must be impure.

While pure functions can't call impure actions, there's no rule to prevent the obverse. Impure actions *can* call pure functions.

Therefore, the best we can ever hope to achieve is an impure entry point that calls pure code and impurely reports the result from the pure function.



L'A et l'Ω du Type Driven Dev...

Le Pattern Matching

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Exemple avec un enum tout bête :

```
pub enum LoginError {
    AuthenticationError(String),
    SSOErrorCode(u32),
    UnexpectedError(String),
}
```

L'opérateur `match`

```
fn my_function(e: LoginError) {
    match e {}
          +-------------------------------------------------------------------------------+
}         |missing match arm: `AuthenticationError(_)`, `SSOErrorCode(_)` and             |
          |`UnexpectedError(_)` not covered (rust-analyzer E0004)                         |
          |-------------------------------------------------------------------------------|
          |non-exhaustive patterns: `LoginError::AuthenticationError(_)`, `LoginError::   |
          |SSOErrorCode(_)` and `LoginError::UnexpectedError(_)` not covered              |
          |the matched value is of type `LoginError`                                      |
          |                                                                               |
          |Related information:                                                           |
          |                                                                               |
          |  * main.rs#463,10: `LoginError` defined here                                  |
          |  * main.rs#470,12: ensure that all possible cases are being handled by       |
          |adding a match arm with a wildcard pattern, a match arm with multiple or-      |
          |patterns as shown, or multiple match arms: ` {                                 |
          |        LoginError::AuthenticationError(_) | LoginError::SSOErrorCode(_) |     |
          |LoginError::UnexpectedError(_) => todo!(),                                     |
          |    }`                                                                         |
          |                                                                               |
          | (rustc E0004)                                                                 |
          |-------------------------------------------------------------------------------|
          |https://doc.rust-lang.org/error-index.html#E0004                               |
          +-------------------------------------------------------------------------------+
```

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Fort heureusement on a ce qu'il faut avec le LSP de Rust 😌

```
fn my_function(e: LoginError) {
    match e {}
}           +Choose action------+
            |1. Fill match arms |
            +-------------------+
```

Et y a plus qu'à remplir !

```
fn my_function(e: LoginError) {
    match e {
        LoginError::AuthenticationError(_) => todo!(),
        LoginError::SSOErrorCode(_) => todo!(),
        LoginError::UnexpectedError(_) => todo!(),
    }
}
```

On peut laisser la vraie implem' pour plus tard

```
fn my_function(e: LoginError) {
    match e {
        LoginError::AuthenticationError(_) => todo!("Ah, pas cool 😱"),
        LoginError::SSOErrorCode(_) => todo!("Oulà ! Encore moins cool 😓"),
        LoginError::UnexpectedError(_) => todo!("Alors là jpp, moi… 😅"),
    }
}
```

On peut être arbitrairement complexe dans le matching

```
fn my_function(e: LoginError) {
    match e {
        LoginError::AuthenticationError(auth)
          if auth.contains("network") => todo!("Ah, pas cool 😱")
        LoginError::SSOErrorCode(c)
          if c > 42 && c < 1337 => todo!("Oulà ! Encore moins cool 😓"),
        LoginError::UnexpectedError(error)
          if error.is_empty() => todo!("Alors là jpp, moi… 😅"),
    }
}
```

Mais le compilateur veille...

```
fn my_function(e: LoginError) {
    match e {
          +-------------------------------------------------------------------------------+
        Lo|missing match arm: `AuthenticationError(_)` not covered (rust-analyzer E0004) |
          |-------------------------------------------------------------------------------|
        } |non-exhaustive patterns: `LoginError::AuthenticationError(_)` not covered      |
        Lo|the matched value is of type `LoginError`                                      |
        Lo|                                                                               |
    }     |Related information:                                                           |
}         |                                                                               |
          |  * main.rs#463,10: `LoginError` defined here                                  |
          |  * main.rs#475,76: ensure that all possible cases are being handled by       |
          |adding a match arm with a wildcard pattern or an explicit pattern as shown: `,|
          |        LoginError::AuthenticationError(_) => todo!()`                         |
          |                                                                               |
          | (rustc E0004)                                                                 |
          |-------------------------------------------------------------------------------|
          |https://doc.rust-lang.org/error-index.html#E0004                               |
          +-------------------------------------------------------------------------------+
```

Voyez-vous le problème, ici ?

```rust
pub enum Direction {
    North,
    South,
    East,
    West,
}
type Coordinates<'a> = (&'a u32, &'a u32);

pub fn move_player((&x, &y): Coordinates, direction: &Direction) {
    match ((x, y), direction) {
        ((0, 0), _) => todo!("Origine du plan, rien à faire"),
        ((13, 10), Direction::South) => todo!("Ah non y a l'océan par-là !"),
    }
}
```

Le compilateur est plus vigilant que vous 😸

```
pub fn move_player((&x: u32, &y: u32): Coordinates, direction: &Direction) {
    match ((x, y), direction) {
          +-----------------------------------------------------------------------------+
        (((|non-exhaustive patterns: `((1_u32..=12_u32, _), _)` and `((14_u32..=u32::MAX,|
        (((|_), _)` not covered                                                         |
    }     |the matched value is of type `((u32, u32), &Direction)`                     |
}         |                                                                             |
          |Related information:                                                         |
          |                                                                             |
          |  * main.rs#412,77: ensure that all possible cases are being handled by      |
          |adding a match arm with a wildcard pattern, a match arm with multiple or-    |
          |patterns as shown, or multiple match arms: `,                                |
          |        ((1_u32..=12_u32, _), _) | ((14_u32..=u32::MAX, _), _) => todo!()`   |
          |                                                                             |
          | (rustc E0004)                                                               |
          |-----------------------------------------------------------------------------|
          |https://doc.rust-lang.org/error-index.html#E0004                             |
          +-----------------------------------------------------------------------------+
```

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Et finalement, tout s'arrange 😉

```rust
pub enum Direction {
    North,
    South,
    East,
    West,
}
type Coordinates<'a> = (&'a u32, &'a u32);

pub fn move_player((&x, &y): Coordinates, direction: &Direction) {
    match ((x, y), direction) {
        ((0, 0), _) => todo!("Origine du plan, rien à faire"),
        ((13, 10), Direction::South) => todo!("Ah non y a l'océan par-là !"),
        ((_,_) , _) => todo!("Autres cas à traiter plus tard")
    }
}
```

Les "Boîtes" de Rust

Moment Kahoot !

Hé oui, c'est bien Null:

Speaking at a software conference in 2009, Tony Hoare hyperbolically apologized f
[26][27]

> I call it my billion-dollar mistake. It was the invention of the null reference in 1
> designing the first comprehensive type system for references in an object orie
> My goal was to ensure that all use of references should be absolutely safe, wit
> automatically by the compiler. But I couldn't resist the temptation to put in a n
> because it was so easy to implement. This has led to innumerable errors, vuln
> crashes, which have probably caused a billion dollars of pain and damage in th

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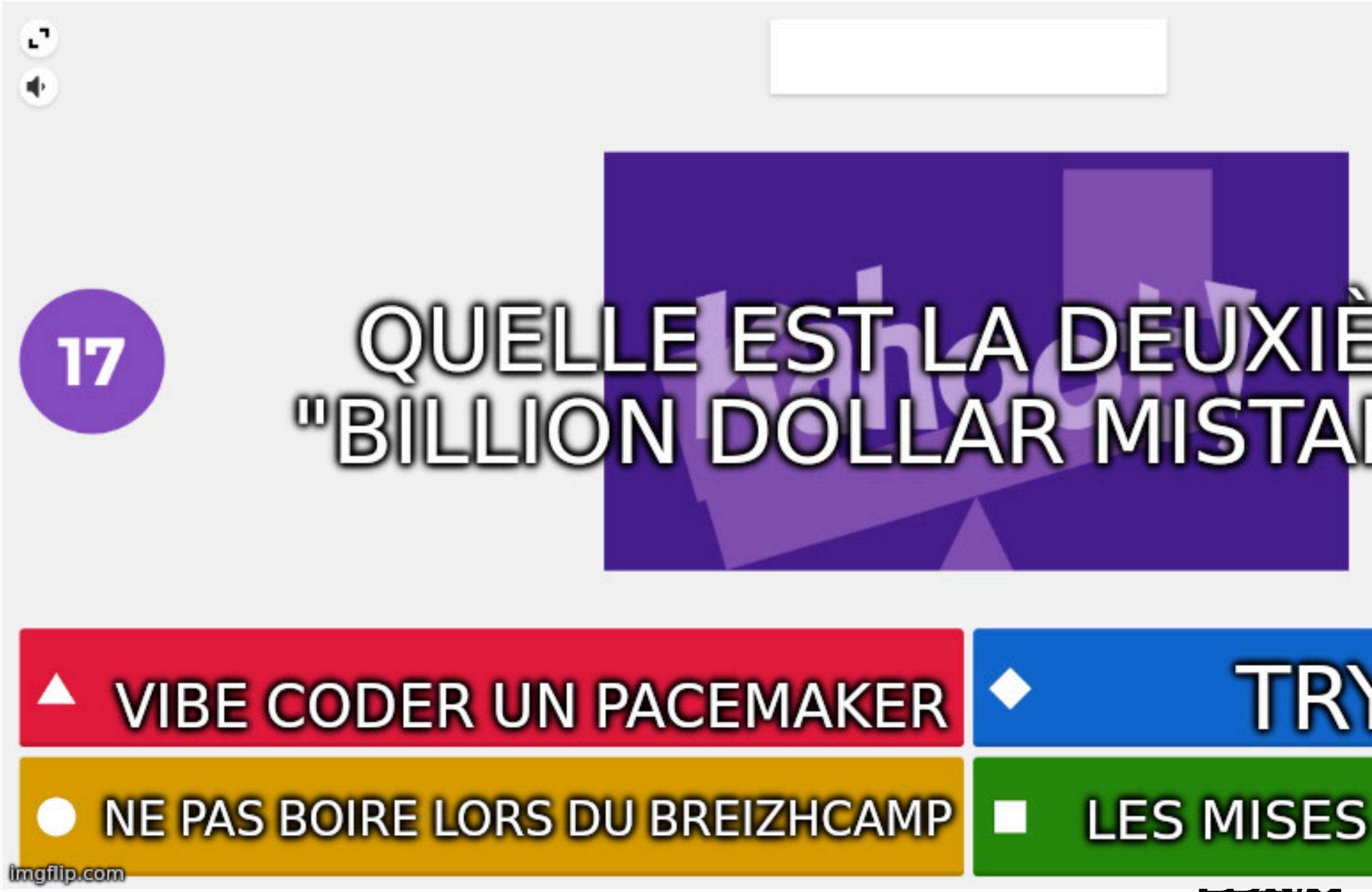
En Rust, plus de doute !

```
// get_user_by_email:
//
//   UserEmail -> Option<User>
//
match get_user_by_email(user_email).await {

    Some(user) => todo!("rajouter le cas où l'utilisateur existe"),

    None => todo!("créer le user ? retourner une erreur ? vous décidez !"),

}
```

Moment Kahoot 2 (Le Retour !)



Why Do Some Developers Consi Problematic?

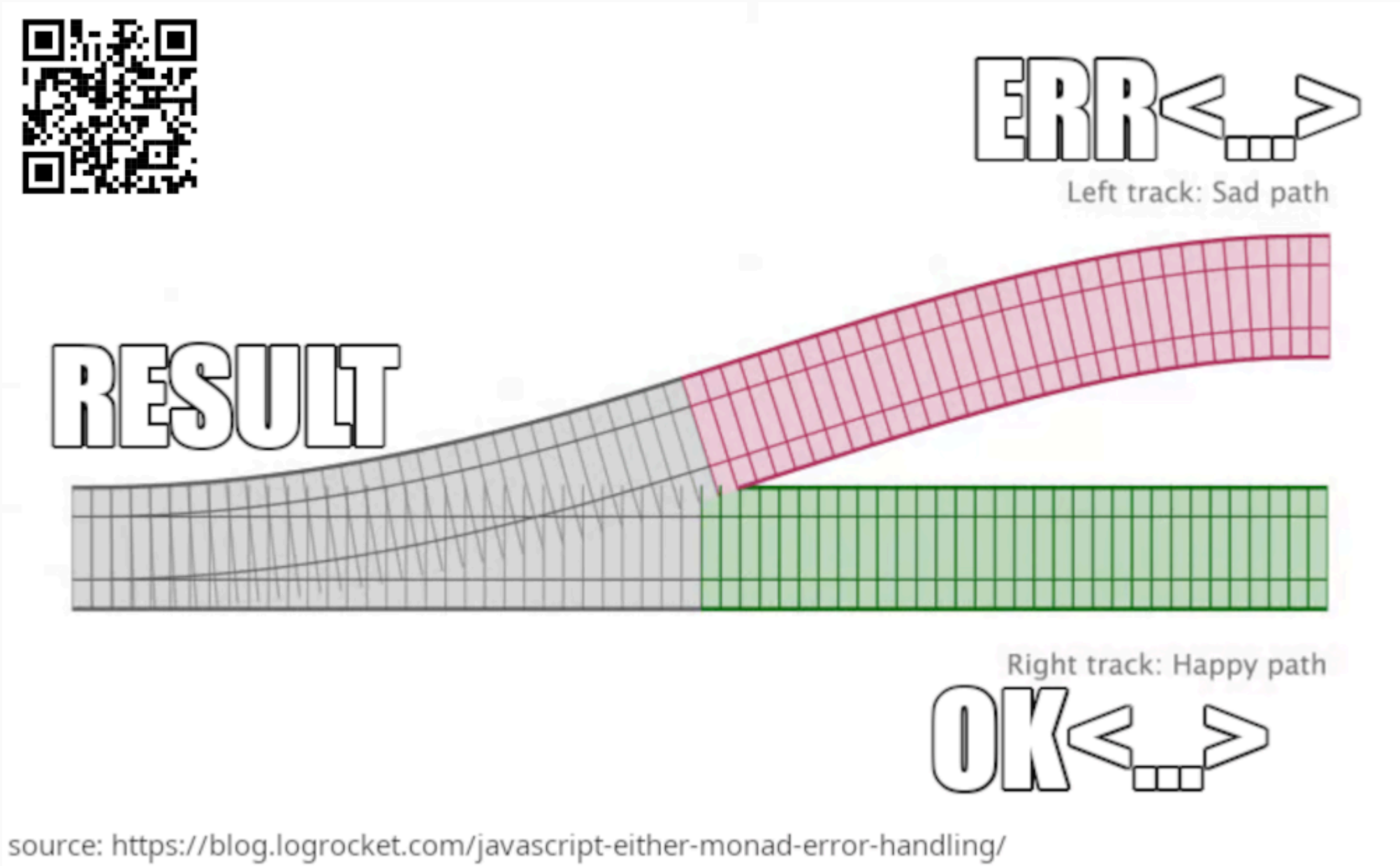
The sentiment that `try-catch` is "evil" stems from several potential pitfall

- **Masking Errors**: Catching exceptions too broadly can hide underlying prob
- **Performance Overhead**: Exception handling can be computationally expen and caught.
- **Code Complexity**: Excessive `try-catch` blocks can clutter code and make
- **Control Flow**: Using exceptions for normal control flow is generally bad pra

Certains langages font ce qu'ils peuvent... 😂

DX I/O

D'autres, comme Rust, plongent dans le multivers ! 💪



source: https://blog.logrocket.com/javascript-either-monad-error-handling/

Rappelez-vous `UserEmail` ! 🤓

```rust
use validator::ValidateEmail;

#[derive(Debug)]
pub struct UserEmail(String);

impl UserEmail {                +-------------------------+
    pub fn parse(s: String) -> |Result<UserEmail, String>| {
                                +-------------------------+
        if s.validate_email() {
           +-----------+
           |Ok(Self(s))|
           +-----------+
        } else {
           +-----------------------------------------------+
           |Err(format!("{} is not a valid user email", s))|
           +-----------------------------------------------+
        }
    }
}
```

De plus, Un Result étant un Enum, on peut le `match` 🤩

```
// validate_credentials:
//
//   Credentials -> Result<Uuid, LoginError>
//
match validate_credentials(credentials).await {
}                       +Choose action -----+
                        |1. Fill match arms |
                        +-------------------+
```

Et y a plus qu'à remplir, ou à `todo` ! 🤷

```
// validate_credentials:
//
//   Credentials -> Result<Uuid, LoginError>
//
match validate_credentials(credentials).await {

    Ok(_) => todo!("rajouter le cas où tout est ok"),

    Err(_) => todo!("rajouter la gestion d'erreur"),

}
```

Finalement, on met le code final:

```
match validate_credentials(credentials).await {
  Ok(user_id) => {
    return HttpResponse::SeeOther()
      .insert_header((LOCATION, "/admin/dashboard"))
      .finish());
  }
  Err(e) => {
    return login_redirect(LoginError::AuthError(e));
  }
}
```

"On est obligés de `match` à chaque fois un Result ?" 🤔

Non, on peut utiliser (entre autres) `.unwrap()`, pour ça:

```rust
// Dans ce cas-ci, on est persuadés qu'on
// aura toujours un port disponible
let port = listener.local_addr().unwrap().port();
```

Mais pourquoi on s'inflige `match`, alors ? ☝️🤓

...et même les plus gros se font avoir 💥

When the bad file with more than 200 features was propagated to our servers, this limit was hit — resulting in the system panicking. The FL2 R code that makes the check and was the source of the unhandled error is shown below:

```
71      /// Fetch edge features based on `input` struct into [`Features`] buffe
72      pub fn fetch_features(
73          &mut self,
74          input: &dyn BotsInput,
75          features: &mut Features,
76      ) -> Result<(), (ErrorFlags, i32)> {
77          // update features checksum (lower 32 bits) and copy edge feature n
78          features.checksum &= 0xFFFF_FFFF_0000_0000;
79          features.checksum |= u64::from(self.config.checksum);
80          let (feature_values, _) = features
81              .append_with_names(&self.config.feature_names)
82              .unwrap();
```

This resulted in the following panic which in turn resulted in a 5xx error:

```
thread fl2_worker_thread panicked: called Result::unwrap() on an Err value
```

https://blog.cloudflare.com/18-november-2025-outage/

Réservez `.unwrap()` pour le code de test et apprenez à propager ou gérer correctement les `Result`

```
fn my_function() -> Result<PortNumber, Error> {
  /* In this case, we SHOULD have an IP address. */
  /* If not, bubble up the error upwards */
  let port = listener.local_addr()?.port();
  return port;
}
```

What it does

Checks for `.unwrap()` or `.unwrap_err()` calls on `Resu`

Why restrict this?

It is better to handle the `None` or `Err` case, or at least c
message. Still, for a lot of quick-and-dirty code, `unwrap`
lint is `Allow` by default.

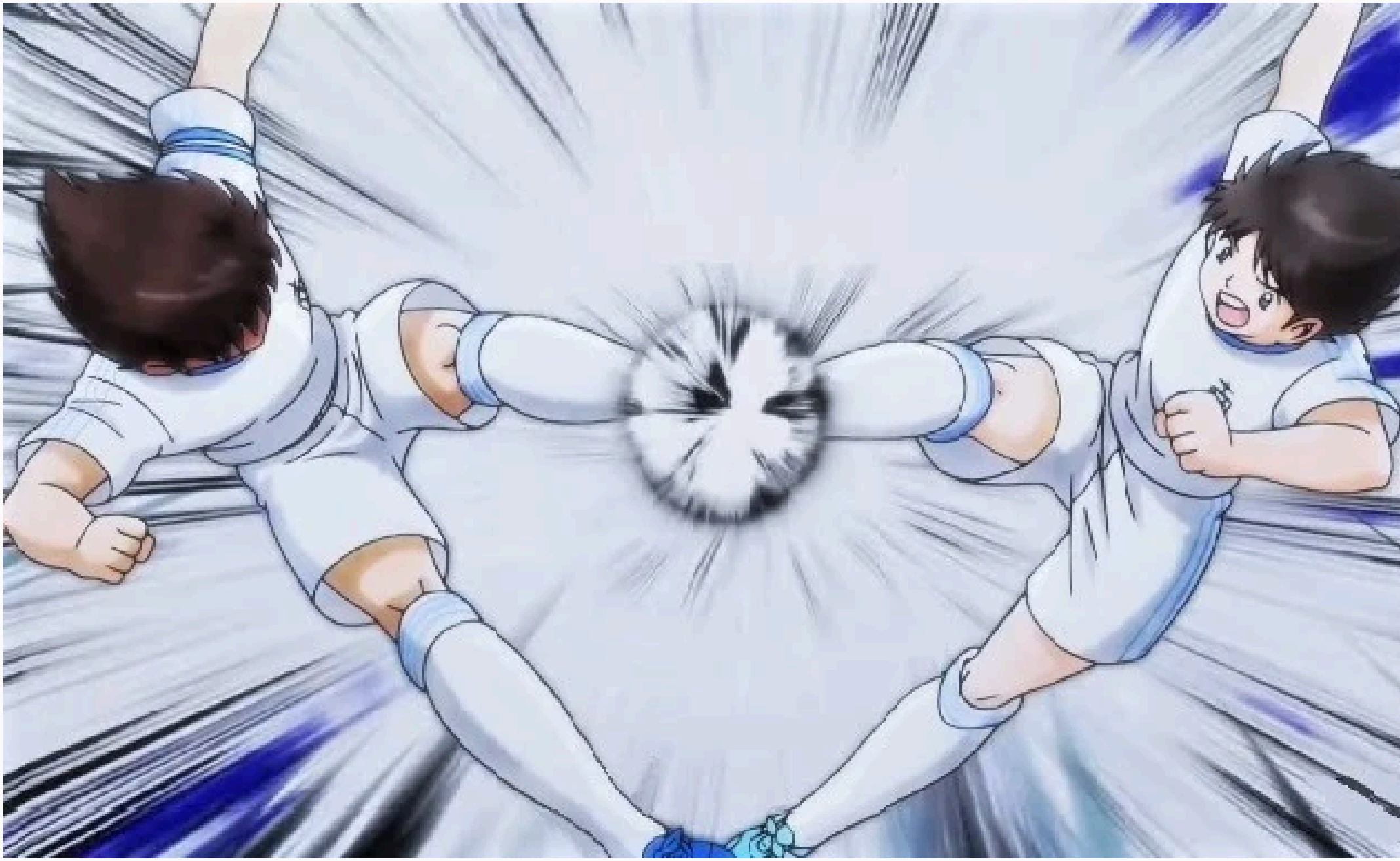
`result.unwrap()` will let the thread panic on `Err` value
more sophisticated error handling, and propagate error

Even if you want to panic on errors, not all `Errors` impl

D'ailleurs, en parlant d'erreurs...

Les jumeaux surdoués de la gestion d'erreur

anyhow et thiserror



anyhow, les erreurs faciles à créer :

```rust
use anyhow::{Context, Result};

/*  Le Result ne précise même plus le type d'erreur */
/*  car ce sera un `anyhow::Error` 👌 */
fn main() -> Result<()> {

  /*  .context permet d'emballer l'erreur avec…du contexte 😁 */
  it.detach().context("Failed to detach the thing")?;

  /*  L'utilisation de `?` veut simplement dire: */
  /*  Si c'est une `Err` alors on bubble up ! */
  /*  Sinon on "ouvre" la boîte `Ok` et on prend son contenu */
  let content = std::fs::read(path)?;
}
```

`thiserror`, les erreurs faciles à décrire :

```rust
use thiserror::Error;
#[derive(Error, Debug)]
pub enum DataStoreError {
  #[error("data store disconnected")]
  Disconnect(#[from] io::Error),
  #[error("the data for key `{0}` is not available")]
  Redaction(String),
  #[error("invalid ({expected:?}, found {found:?})")]
  InvalidHeader {
      expected: String,
      found: String,
  },
  #[error("unknown data store error")]
  Unknown
}
```

`thiserror` pour tous les types ? `derive_more` !

```rust
use derive_more::{Add, Display, From, Into};

#[derive(PartialEq, From, Add)]
struct MyInt(i32);

#[derive(PartialEq, From, Into)]
struct Point2D {
    x: i32,
    y: i32,
}

#[derive(PartialEq, From, Add, Display)]
enum MyEnum {
    #[display("int: {_0}")]
    Int(i32),
    Uint(u32),
    #[display("nothing")]
    Nothing,
}
```

Vous trouvez ça lourd de devoir déclarer un enum ? eros !

```rust
use eros::IntoUnion;
use std::{io, num::ParseIntError};

fn load_port(path: &str) -> eros::Result<u16, (io::Error, ParseIntError)> {
    let contents = std::fs::read_to_string(path).union()?;
    contents.trim().parse().union()
}

fn port_or_default(path: &str) -> eros::Result<u16, (ParseIntError,)> {
    load_port(path).recover::<io::Error, _>(|error| {
        eprintln!("{error:?}; using port 8080");
        8080
    })
}
```

Traits, la POO turbo-chargée

On définit ce qu'est un PaymentProcessor

```rust
pub struct RecipientEmail(pub String);
pub struct SenderEmail(pub String);

pub trait PaymentProcessor {
    fn send_payment(
        recipient: &RecipientEmail,
        sender: &SenderEmail,
    ) -> Result<(), anyhow::Error>;
}
```

On veut ensuite implémenter "notre" PaymentProcessor

```rust
pub struct RecipientEmail(pub String);
pub struct SenderEmail(pub String);

pub trait PaymentProcessor {
    fn send_payment(
        recipient: &RecipientEmail,
        sender: &SenderEmail,
    ) -> Result<(), anyhow::Error>;
}

pub struct MyPaymentProcessor {}

impl PaymentProcessor for MyPaymentProcessor {}
```

```rust
pub struct RecipientEmail(pub String);
pub struct SenderEmail(pub String);

pub trait PaymentProcessor {
    fn send_payment(
        recipient: &RecipientEmail,
        sender: &SenderEmail,
    ) -> Result<(), anyhow::Error>;
}

pub struct MyPaymentProcessor {}

impl PaymentProcessor for MyPaymentProcessor {}
```

```
   +-----------------------------------------------------------------------------+
   |not all trait items implemented, missing: `send_payment`                    |
   |missing `send_payment` in implementation                                     |
   |                                                                             |
   |Related information:                                                         |
   |                                                                             |
   |  * main.rs#422,5: `send_payment` from trait                                 |
   |  * main.rs#431,47: implement the missing item: `fn send_payment(_:          |
   |&RecipientEmail, _: &SenderEmail) -> std::result::Result<(), anyhow::Error> { |
   |todo!() }`: `fn send_payment(_: &RecipientEmail, _: &SenderEmail) ->         |
   |std::result::Result<(), anyhow::Error> { todo!() }`                          |
   +-----------------------------------------------------------------------------+
```

Comme pour les enums, on peut utiliser le puissant LSP de Rust

```
pub struct MyPaymentProcessor {}

impl PaymentProcessor for MyPaymentProcessor {}
      +Choose action --------------+
      |1. Implement missing members|
      +----------------------------+
```

Et on met ensuite l'implem' qu'on veut:

```rust
pub struct RecipientEmail(pub String);
pub struct SenderEmail(pub String);

pub trait PaymentProcessor {
    fn send_payment(
        recipient: &RecipientEmail,
        sender: &SenderEmail,
    ) -> Result<(), anyhow::Error>;
}

pub struct MyPaymentProcessor {}
impl PaymentProcessor for MyPaymentProcessor {
    fn send_payment(
        recipient: &RecipientEmail,
        sender: &SenderEmail,
    ) -> Result<(), anyhow::Error> {
        todo!("Payment instructions from {sender} to {recipient} go here !")
    }
}
```

Les traits "standards"

Un exemple simple: J'ai une `LoginError`, et je veux la déboguer

```
pub enum LoginError {
    AuthenticationError(String),
    UnexpectedError(String),
}
```

Une "fonction" super pratique: dbg !

```
let my_error = LoginError::AuthError("Oulà !");
dbg!(&my_error)
```

Sauf que LoginError n'est pas un type élémentaire...

```
+-------------------------------------------------------------+
|`LoginError` doesn't implement `std::fmt::Debug`             |
|add `#[derive(Debug)]` to `LoginError` or manually `impl     |
|std::fmt::Debug for LoginError`                              |
|                                                             |
|Related information:                                         |
|                                                             |
|  * error.rs#53,18: required by a bound in `Error`           |
|  * post.rs#18,1: consider annotating `LoginError` with      |
|    `#[derive(Debug)]`                                       |
|                                                             |
| (rustc E0277)                                               |
|-------------------------------------------------------------|
|https://doc.rust-lang.org/error-index.html#E0277             |
+-------------------------------------------------------------+
```

Dériver, simple et bien pratique dans 90% des cas :

```
#[derive(Debug)]
pub enum LoginError {
    AuthError(Error),
    UnexpectedError(Error),
}
```

Ok pour les cas de base, mais comment implémenter explicitement Debug ?

Encore une fois, le LSP est là pour nous aider 😍

```
pub enum LoginError {
    AuthError(String),
    UnexpectedError(String),
}

impl std::fmt::Debug for LoginError {}
              +Choose action --------------+
              |1. Implement missing members|
              +----------------------------+
```

On obtient une implémentation "par defaut", qu'on peut alors customiser 🤤

```rust
impl std::fmt::Debug for LoginError {
  fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
    match self {
      Self::AuthError(arg0) =>
          f.debug_tuple("AuthError").field(arg0).finish(),
      Self::UnexpectedError(arg0) =>
          f.debug_tuple("UnexpectedError").field(arg0).finish(),
    }
  }
}
```

Et on y met ce qu'on veut (tant qu'on respecte le Trait)

```
impl std::fmt::Debug for LoginError {
    fn fmt(&self, f: &mut Formatter<'_>) -> Result {
        writeln!(f, "{}\n", e)?;
        let mut current = e.source();
        while let Some(cause) = current {
            writeln!(f, "Caused by:\n\t{}", cause)?;
            current = cause.source();
        }
        Ok(())
    }
}
```

Ce qui nous permet d'avoir un détail de l'erreur bien plus clair en debug :

```
Failed to log in user.
Caused by:
    Failed to retrieve user informations
Caused by:
    error returned from database
Caused by:
    table "user" does not exist
```

BDX I/O

Idée: je veux customiser la manière de loguer de mes variables 🤔

```rust
trait Loggable { fn log(&self); }

pub struct Cat { pub name: String, }

fn main() {
    let numbers = vec![1, 2, 3];
    numbers.log(); // Erreur ! 💥
    // Ah, mais oui ! Faut que j'implément Loggable pour u64 😫

    let s = String::from("Blanquette de Veau");
    s.log(); // Erreur ! 💥
    // Ah, mais oui ! Faut que j'implément Loggable pour String 😫

    let c = Cat { name: "Mimouchka".to_string(), };
    c.log(); // Erreur ! 💥
    // Ah, mais oui ! Faut que j'implément Loggable pour Cat 😫
}
```

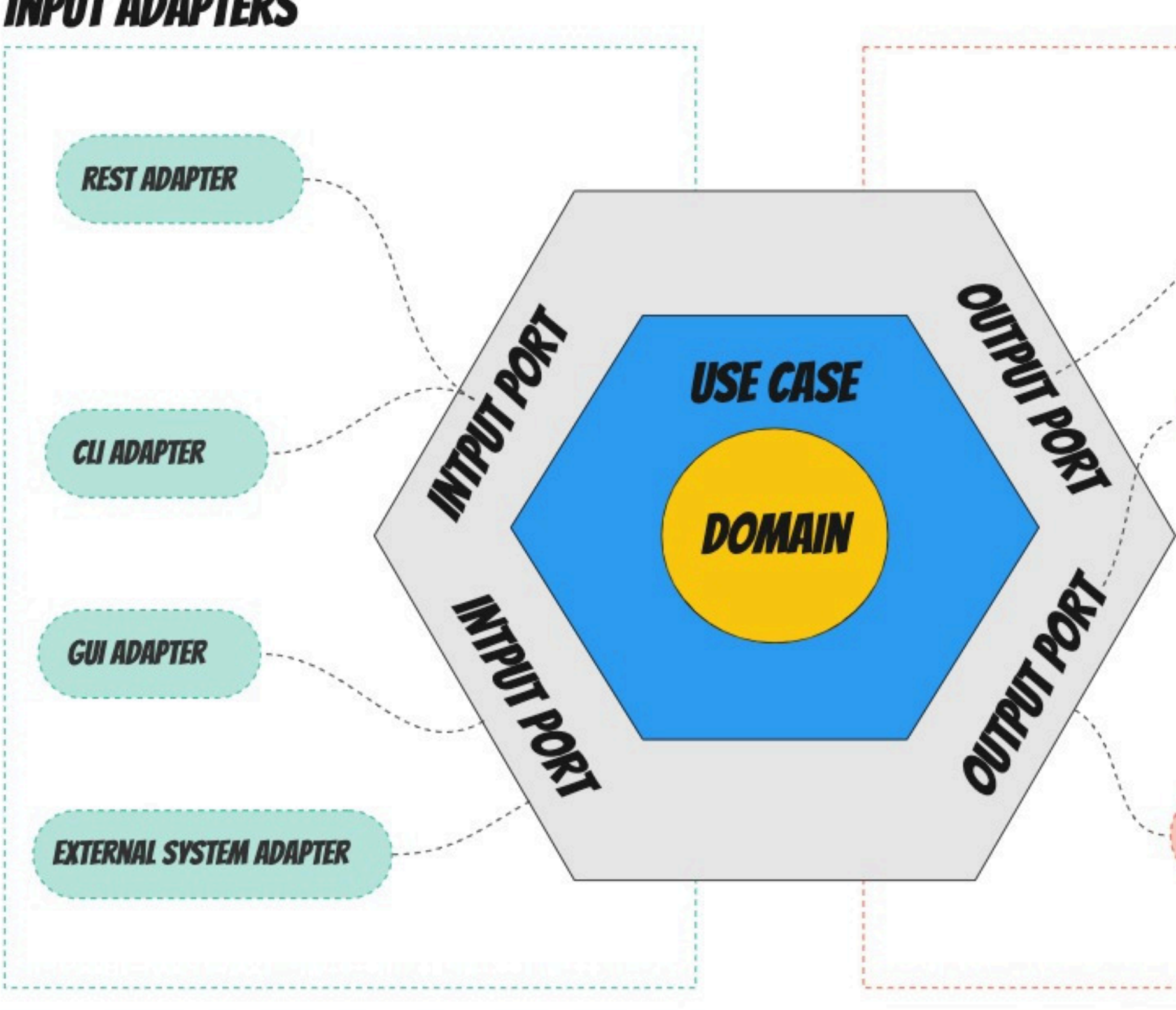
`impl<T>`, parce que nous le valons bien 💇

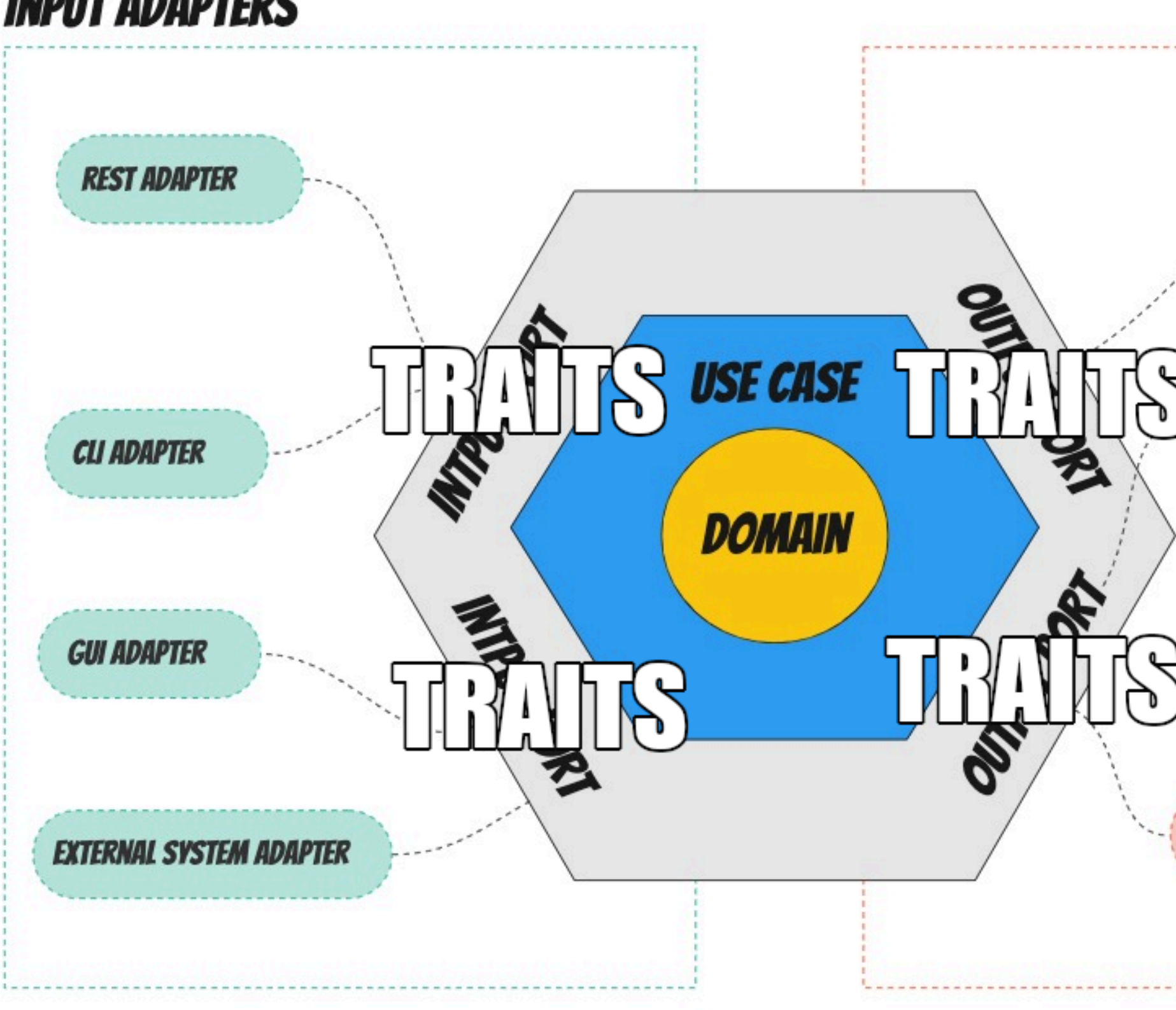
```rust
trait Loggable { fn log(&self); }

// Tout ce qui impl Debug va désormais automatiquement impl Loggable ! 🤩
impl<T> Loggable for T
where T: Debug {
    fn log(&self) { println!("[log] {:?}", self); }
}

#[derive(Debug)]
pub struct Cat { pub name: String, }

fn main() {
    let numbers = vec![1, 2, 3];
    numbers.log(); // Vec<i32> impl nativement Debug, et donc Loggable ! 😌

    let s = String::from("Blanket");
    s.log(); // String impl nativement Debug, et donc Loggable ! 😌

    let c = Cat { name: "Astra".to_string(), };
    c.log(); // Cat impl Debug (via derive), et donc Loggable ! 😌
}
```

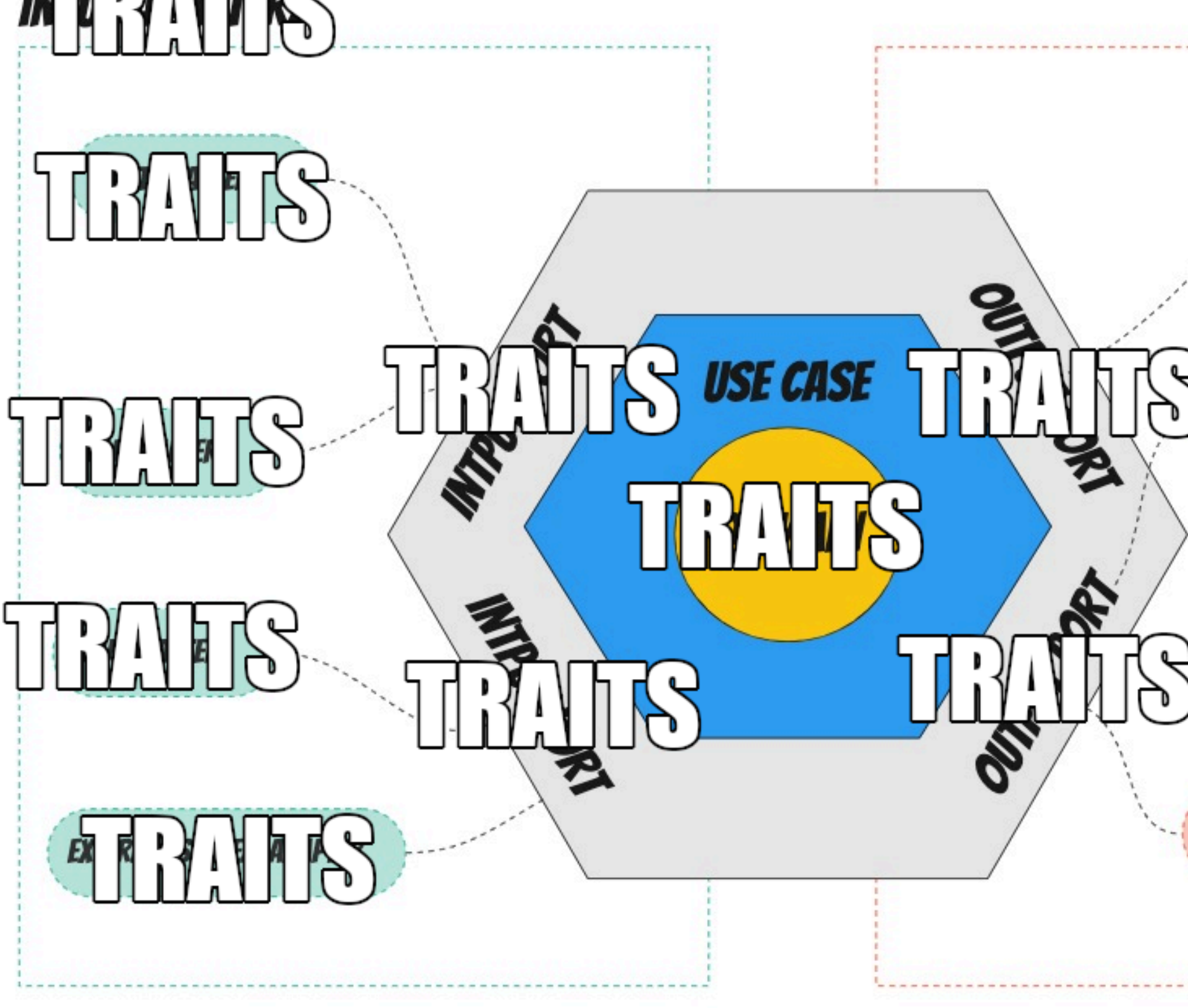
Et des types/traits/impl comme ça, vous en avez PLEIN, dans Rust 🤩

```
- `Option` (`Some<…>` ou `None`)
- `Debug` vs `Display`
- `Deref`
- `Drop`
- ...et plein d'autres (sans compter ceux de la communauté 🛡)
```

HEXAGON
WHAT'S THE SQUARE ROOT OF 965?
31.0644
DOES MY DAD LOVE ME?
YES
Pourquoi Rust serait sûr alors qu'il y a littéralement un mot-clé "unsafe" ?


INPUT ADAPTERS
REST ADAPTER
CLI ADAPTER
GUI ADAPTER
EXTERNAL SYSTEM ADAPTER
INTPUT PORT
INTPUT PORT
USE CASE
DOMAIN
OUTPUT PORT
OUTPUT PORT


INPUT ADAPTERS
REST ADAPTER
CLI ADAPTER
GUI ADAPTER
EXTERNAL SYSTEM ADAPTER
TRAITS
USE CASE
TRAITS
DOMAIN
TRAITS
TRAITS


TRAITS
TRAITS
TRAITS
TRAITS
TRAITS
TRAITS
TRAITS
TRAITS
USE CASE
TRAITS
TRAITS
TRAITS
TRAITS

Et c'est ok !

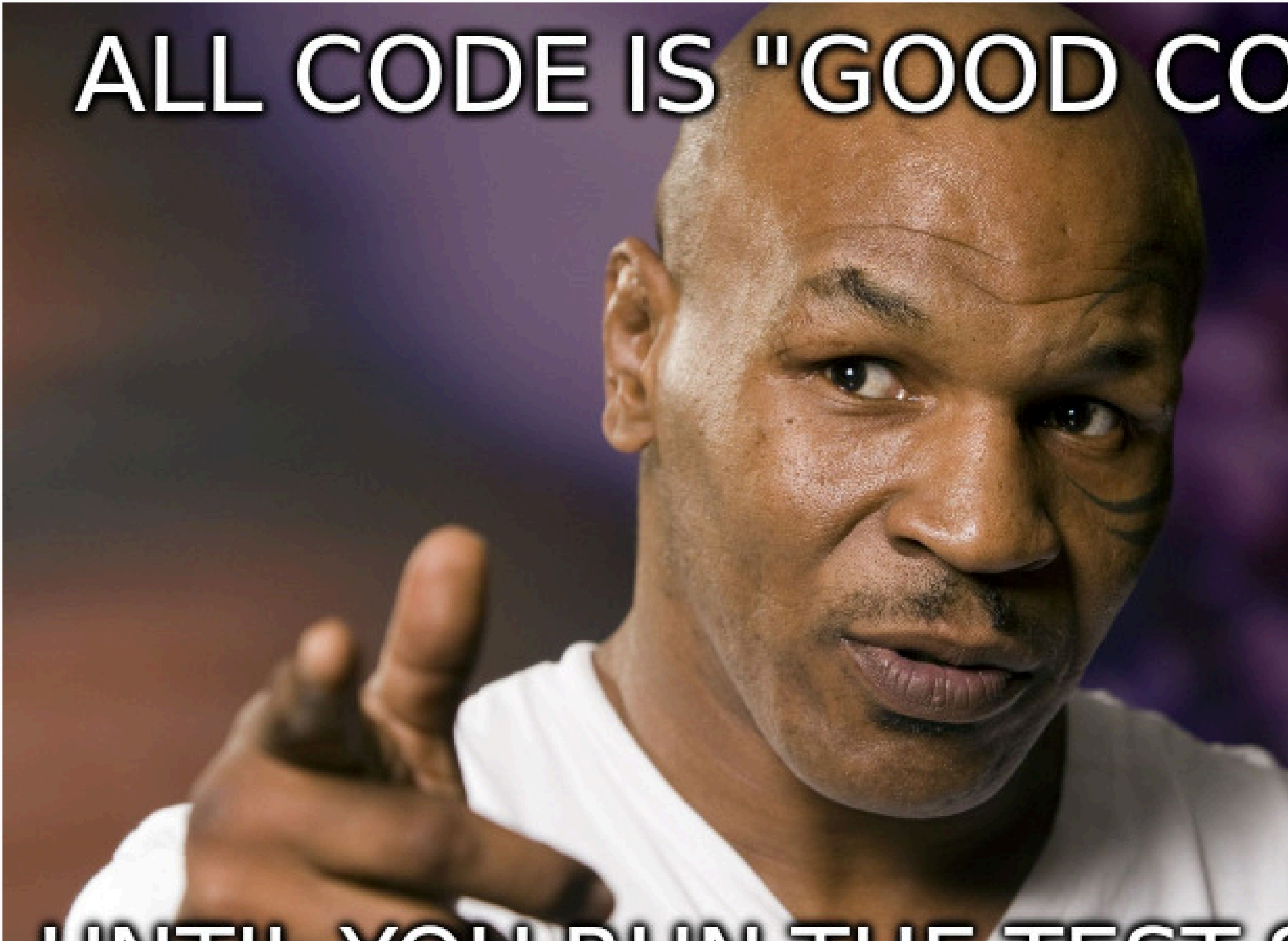
```rust
fn process_data<F, B>(foo: F, bar: B) -> String
where
    F: Clone + Debug,
    B: Display + PartialEq,
{
    // Function implementation
    format!("{:?} - {}", foo, bar)
}
```

```rust
pub async fn create_author<AR: AuthorRepository>(
    State(state): State<AppState<AR>>,
    Json(body): Json<CreateAuthorHttpRequestBody>,
) -> Result<ApiSuccess<CreateAuthorResponseData>, ApiError> {
    let domain_req = body.try_into_domain()?;
    state
        .author_repo
        .create_author(&domain_req)
        .await
        .map_err(ApiError::from)
        .map(|ref author| ApiSuccess::new(StatusCode::CREATED, aut
}
```

source: https://www.h

Oh my.

Isn't it beautiful?


ALL CODE IS "GOOD CO

```
Ω …/temp-rust  trunk  v1.93.0 ?  Φ
(ins)› cargo test -q

running 2 tests
. 1/2
tests::test_adder_ko ── FAILED

failures:

─── tests::test_adder_ko stdout ───

thread 'tests::test_adder_ko' (186854) panicked at src/lib.rs:23:9:
assertion `left == right` failed
  left: 3
 right: 4
note: run with `RUST_BACKTRACE=1` environment variable to display a backtrace


failures:
    tests::test_adder_ko

test result: FAILED. 1 passed; 1 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.0
0s

error: test failed, to rerun pass `--lib`
```

```
10 /// Adder: Add tw
 9 /// ```
 8 /// let x = 5;
 7 /// let y = 5;
 6 /// assert_eq!(my
 5 /// ```
 4 pub fn adder<T>(x
 3 where T: std::ops
 2     x + y
 1 }
 0
 1 #[cfg(test)]
 2 mod tests {
 3     use super::*;
 4
 5     #[test]
 6     fn test_adder
 7         assert_eq
 8     }
 9
10     #[test]
11     fn test_adder
12         assert_eq
13     }
14 }
15
```

```
use mylib::adder;

fn main() {
println!("{}", adder(40, 2));
}
```

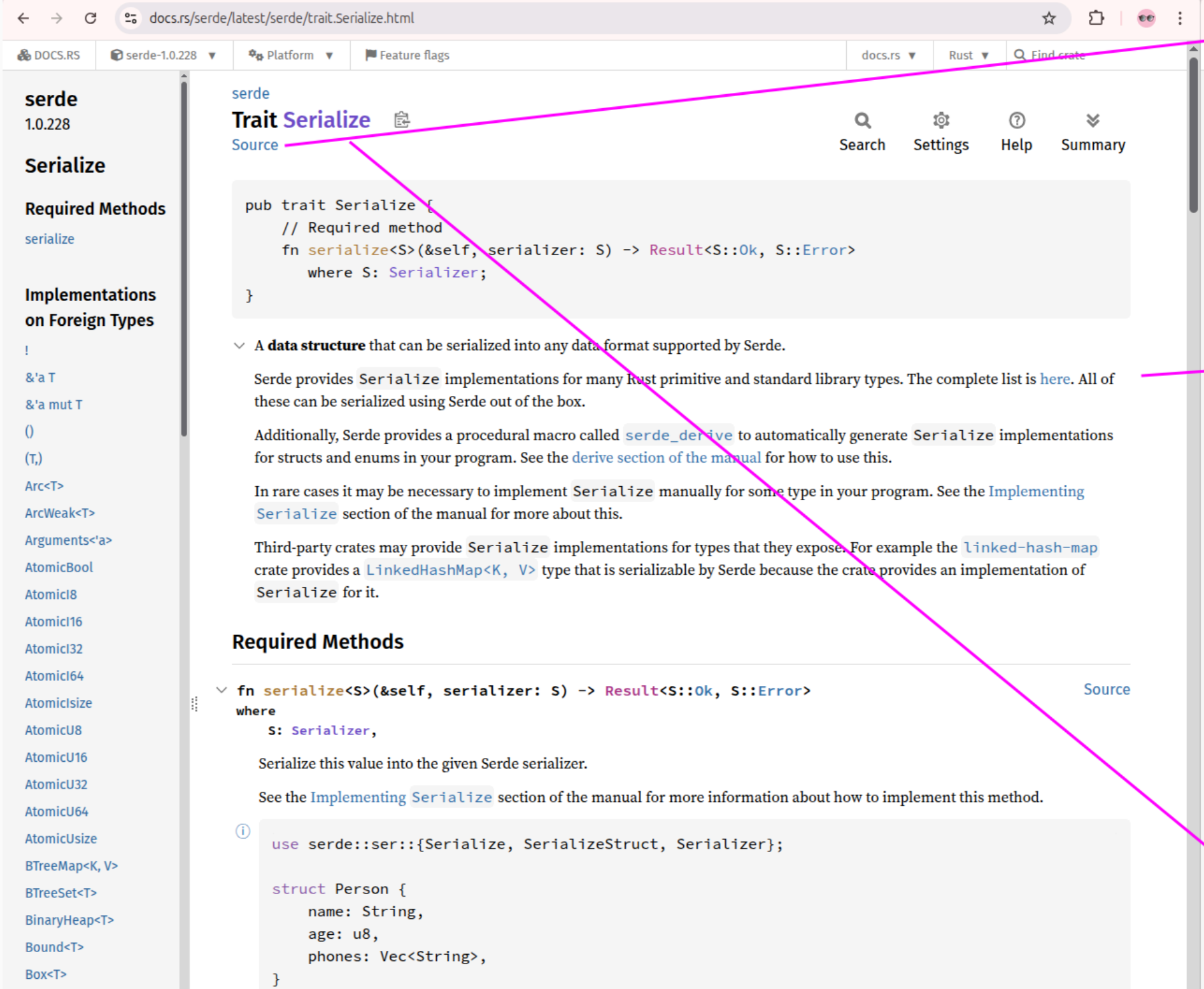
mylib

```
pub fn adder<T>(x: T, y: T)
where
    T: std::ops::Add<Output
```

T = i32

Adder: Add two numbers.

```
let x = 5;
let y = 5;
assert_eq!(mylib::adder(x,y
```


docs.rs/serde/latest/serde/trait.Serialize.html
DOCS.RS
serde-1.0.228
Platform
Feature flags
docs.rs
Rust
Find crate
serde
1.0.228
Serialize
Required Methods
serialize
Implementations on Foreign Types
!
&'a T
&'a mut T
()
(T,)
Arc<T>
ArcWeak<T>
Arguments<'a>
AtomicBool
AtomicI8
AtomicI16
AtomicI32
AtomicI64
AtomicIsize
AtomicU8
AtomicU16
AtomicU32
AtomicU64
AtomicUsize
BTreeMap<K, V>
BTreeSet<T>
BinaryHeap<T>
Bound<T>
Box<T>
serde
Trait Serialize
Source
Search
Settings
Help
Summary
pub trait Serialize {
// Required method
fn serialize<S>(&self, serializer: S) -> Result<S::Ok, S::Error>
where S: Serializer;
}
A data structure that can be serialized into any data format supported by Serde.
Serde provides Serialize implementations for many Rust primitive and standard library types. The complete list is here. All of these can be serialized using Serde out of the box.
Additionally, Serde provides a procedural macro called serde_derive to automatically generate Serialize implementations for structs and enums in your program. See the derive section of the manual for how to use this.
In rare cases it may be necessary to implement Serialize manually for some type in your program. See the Implementing Serialize section of the manual for more about this.
Third-party crates may provide Serialize implementations for types that they expose. For example the linked-hash-map crate provides a LinkedHashMap<K, V> type that is serializable by Serde because the crate provides an implementation of Serialize for it.
Required Methods
fn serialize<S>(&self, serializer: S) -> Result<S::Ok, S::Error>
where
S: Serializer,
Source
Serialize this value into the given Serde serializer.
See the Implementing Serialize section of the manual for more information about how to implement this method.
use serde::ser::{Serialize, SerializeStruct, Serializer};
struct Person {
name: String,
age: u8,
phones: Vec<String>,
}
DOCS.RS
serde-
195 dec
196
197 //
198
199 //
200 //
201 //
202 //
203 //
204 //
205 //
206 //
207 //
208 //
209 //
210 //
211 //
212 //
213 //
214 //
215 //
216 //
217 //
218 //
219 //
220 //
221 //
222 //
223 //
224 #[
225
226
227
228
229
230
231
232
233)]
234 pub
235

Fichier /home/stephane/projets/perso/temp-rust/target/doc/mylib/fn.adder.html

mylib
0.1.0

Functions

adder

mylib

Function adder

Source

Search Settings Help Summary

```
pub fn adder<T>(x: T, y: T) -> T
where
    T: Add<Output = T>,
```

Adder: Add two numbers.

```
let x = 5;
let y = 5;
assert_eq!(mylib::adder(x,y), 10);
```

```
/// ```
/// let
/// let
/// asse
/// ```
pub fn a
where T
    x +
}

#[cfg(te
mod tes
    use

    #[te
    fn t

    }

    #[te
    fn t

    }
}
```

…/temp-rust trunk v1.93.0 ?

Oh, et ils sont *par défaut* testés par `cargo test` 😎

```
running 1 test
.
test result: ok. 1 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out;


running 0 tests

test result: ok. 0 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out;


running 1 test
.
test result: ok. 1 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out;

all doctests ran in 0.15s; merged doctests compilation took 0.14s
```

```
running 0 tests

test result: ok. 0 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; fi


running 1 test
src/lib.rs - adder (line 2) ... FAILED

failures:

---- src/lib.rs - adder (line 2) stdout ----
Test executable failed (exit status: 101).

stderr:

thread 'main' (196168) panicked at /tmp/rustdoctestQGu8L7/doctest_bundle_2024.
assertion `left == right` failed
  left: 10
 right: 11
note: run with `RUST_BACKTRACE=1` environment variable to display a backtrace



failures:
    src/lib.rs - adder (line 2)
```

Tests d'Intégration ?

Facile ! On les met dans `/tests` (avec du code de setup si besoin)

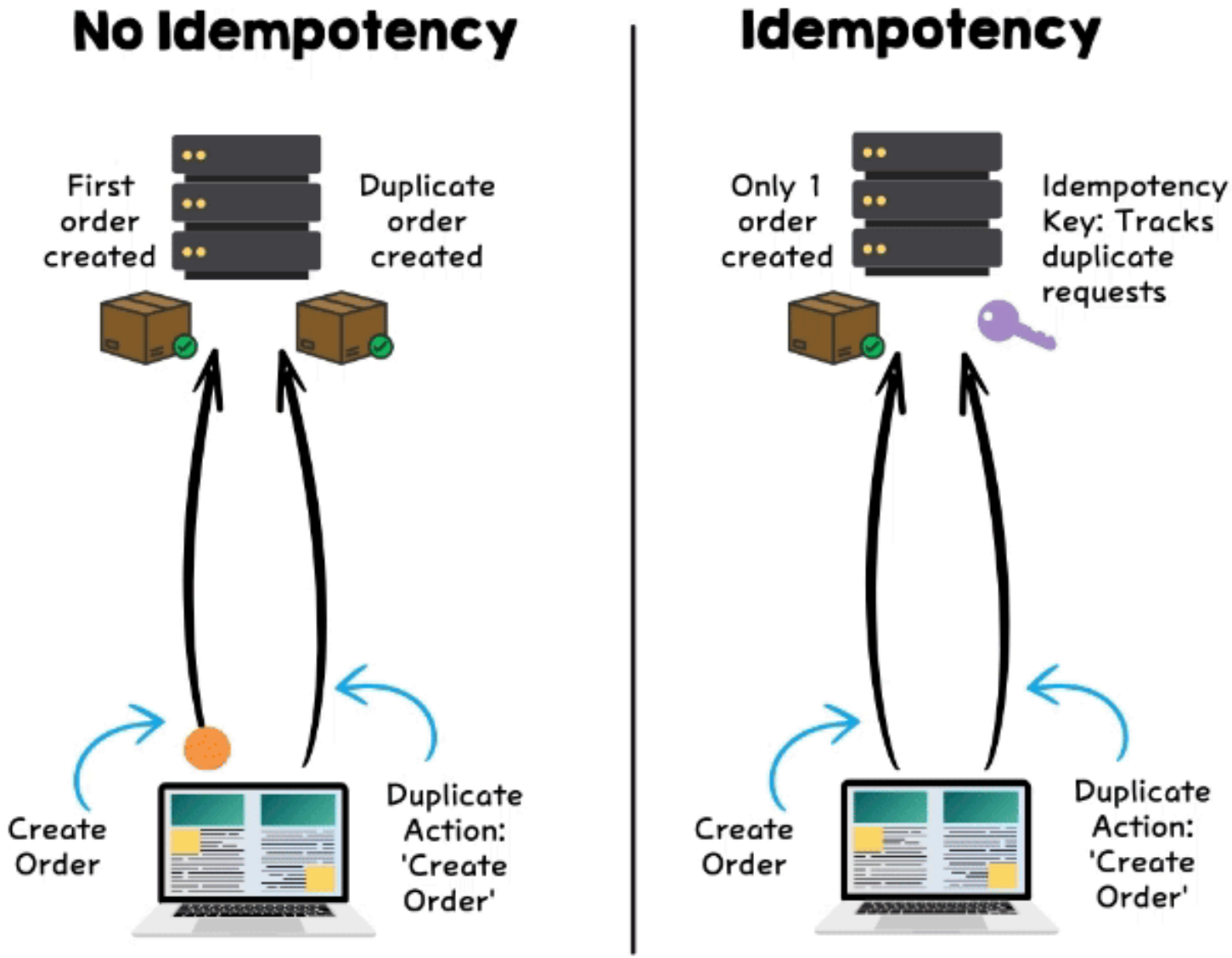
```
0  #[tokio::test]
1  async fn you_must_be_logged_in_to_access_the
2      // Arrange
3      let app: TestApp = spawn_app().await;
4
5      // Act
6      let response: Response = app.get_admin_d
7
8      // Assert
9      assert_is_redirect_to(&response, locatio
10 }
```

NORMAL trunk! tests/api/login.rs rust-analyz

D'ailleurs… Il faut qu'on parle de Wiremocks 🤩

API Design

Sketech newsletter by Nina



Sketech

NinaDurann Sketech

BDX I/O

```
    // Arrange
    let app: TestApp = spawn_app().await;
    create_confirmed_subscriber(&app).await;

    Mock::given(matcher: path("/email")) MockBuilder
        .and(matcher: method("POST")) MockBuilder
        .respond_with(responder: ResponseTemplate::new(2
        // Assert made at Drop
        .expect(1) Mock
        .mount(&app.email_server) impl Future<Output = (
        .await;

    // Act - Part 1 - Login
    app.test_user.login(&app).await;

    // Act - Part 2 - Post new issue
+-- 16 lines: let id = Uuid::new_v4().to_string();·····

    // Act - Part 3 - Post duplicate request
+--  9 lines: assert_is_redirect_to(·····················
```

Ça, c'est le moment où j'ai décidé de me carciniser 🦀

Un écosystème complet

Il faut qu'on parle de sqlx

```
#[tracing::instrument(name = "Retrieve confirmed subscr
pub async fn get_confirmed_subscribers(
    pool: &PgPool,
) → Result<Vec<Result<ConfirmedSubscriber, anyhow::Err
    let confirmed_subscribers = sqlx::query!(
        r#"
        SELECT email from subscriptions WHERE statsu =
        "#,
    type annotations needed
    cannot infer type (rustc E0282)
    ------------------------------------------------------
    https://doc.rust-lang.org/error-index.html#E0282
    ------------------------------------------------------
    error returned from database: column "statsu" does

    Related information:

      * mod.rs#328,9: Actual error occurred here

     (rustc)
```

```rust
#[tracing::instrument(name = "Retrieve presentations from the database", skip(pool))]
pub async fn get_presentations(
    pool: &PgPool,
    product_id: Uuid,
) -> Result<Vec<ProductPresentationSimple>, anyhow::Error> {
    let presentations: Vec<ProductPresentationSimple> = sqlx::query!(
        r#"
        SELECT Q.id, Q.unit_name, Q.unit_size, Q.packaging_name, Q.packaging_factor,
            Q.alternative_unit_name, Q.alternative_unit_factor, Q.created_at, Q.updat
        FROM products_presentations Q
        LEFT OUTER JOIN products_presentations_crosstable X
            ON X.presentation_id = Q.id
        LEFT OUTER JOIN products P
            ON P.id = X.product_id
        WHERE P.id = $1;
        "#,
        product_id
    )
    .fetch_all(pool)
    .await
    .unwrap()
    .into_iter()
    .map(|r| ProductPresentationSimple {
        id: r.id,
        unit_name: r.unit_name,
        unit_size: r.unit_size as u16,
        packaging_name: r.packaging_name,
        packaging_factor: r.packaging_factor as u16,
        alternative_unit_name: r.alternative_unit_name,
        alternative_unit_factor: r.alternative_unit_factor as u16,
        created_at: r.created_at,
        updated_at: r.updated_at,
    })
    .collect();

    Ok(presentations)
```

BDX I/O

Les autres poids lourds :

serde/deser
(sé/désérialisation)
tera (templating HTML)
chrono/jiff (le temps !)
et bien, bien d'autres...

Montons encore d'un cran...

Les Frameworks haut-niveau de Rust 🦀

BDX I/O

Axum

```rust
use axum::{ routing::{get, post}, http::StatusCode, Json, Router, };
use serde::{Deserialize, Serialize};

#[tokio::main]
async fn main() {
    tracing_subscriber::fmt::init(); // initialize tracing
    let app = Router::new()
        .route("/users", post(create_user)); // `POST /users` goes to `create_user`
    let listener = tokio::net::TcpListener::bind("0.0.0.0:3000").await.unwrap();
    axum::serve(listener, app).await;
}

#[derive(Deserialize)]
struct CreateUser { username: String, }

#[derive(Serialize)]
struct User { id: u64, username: String, }

async fn create_user(
    Json(payload): Json<CreateUser>, // Parse the request body as JSON into a `CreateUser` type
) -> (StatusCode, Json<User>) {
    let user = User { id: 1337, username: payload.username, };
    (StatusCode::CREATED, Json(user)) // Converts into a JSON response with 201 HTTP code
}
```

Actix-web

JSON

`Json<T>` allows deserialization of a request body into a struct. To extract typed informat
type `T` must implement `serde::Deserialize`.

```
use actix_web::{post, web, App, HttpServer, Result};
use serde::Deserialize;

#[derive(Deserialize)]
struct Info {
    username: String,
}

/// deserialize `Info` from request's body
#[post("/submit")]
async fn submit(info: web::Json<Info>) -> Result<String> {
    Ok(format!("Welcome {}!", info.username))
}
```

Et le monde des Rustacé-es(c) va vite, très vite… 🏃💨

Introduction

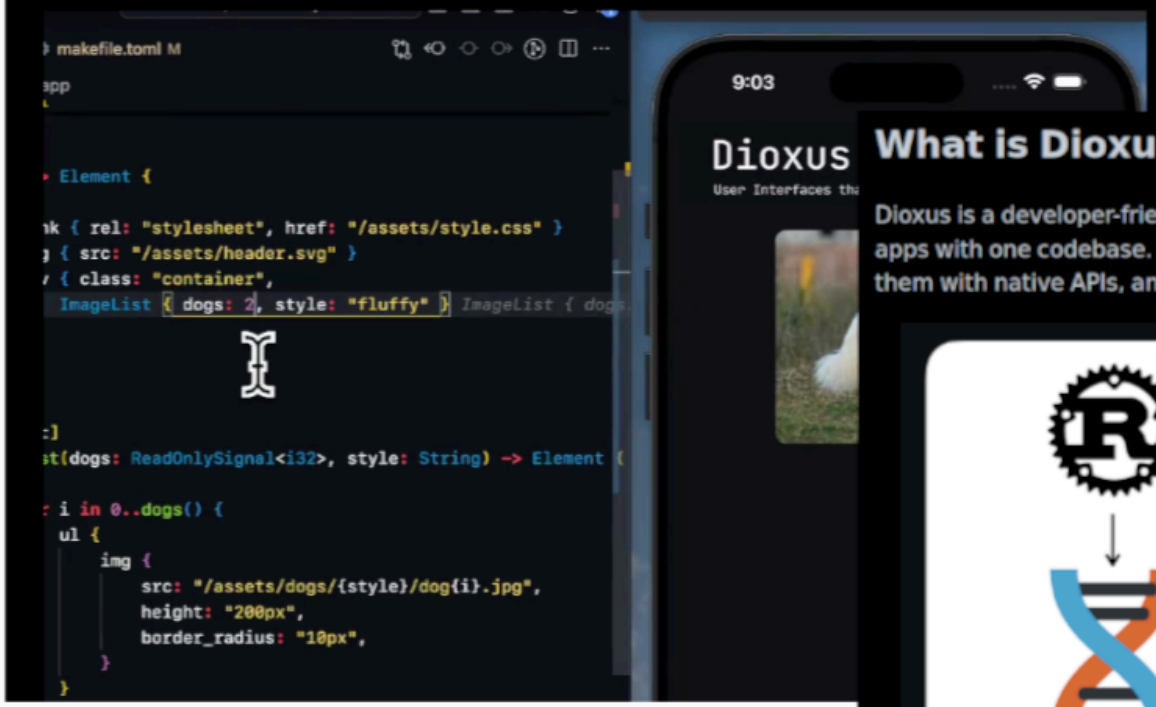
Dioxus is a framework for building cross-platform apps with the Rust progra
language. With one codebase, you can build apps that run on web, desktop,
platforms.

src/readme.rs

```rust
use dioxus::prelude::*;

pub fn App() -> Element {
    let mut count = use_signal(|| 0);

    rsx! {
        h1 { "High-Five counter: {count}" }
        button { onclick: move |_| count += 1, "Up high!" }
        button { onclick: move |_| count -= 1, "Down low!" }
    }
}
```

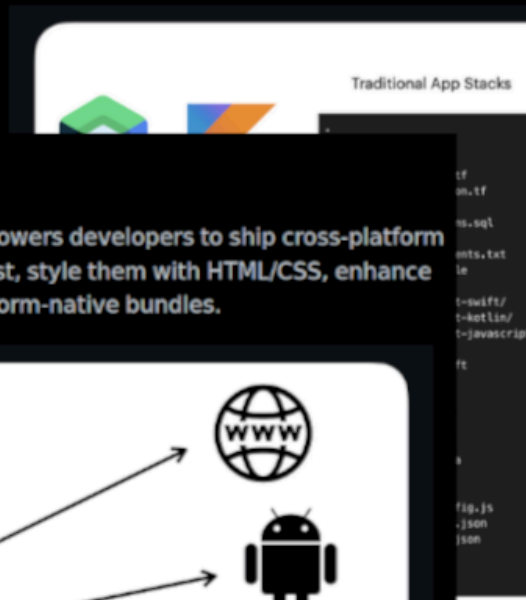
Stellar Developer Experience

With Dioxus, we try to maintain a very high bar for developer experience. We believe that building apps should be fun and straightforward. We've worked to push forward Rust itself, developing technologies like Subsecond Rust hot-reloading, WASM bundle-splitting, linker-based asset bundling, and a modular WGPU-based HTML/CSS renderer.



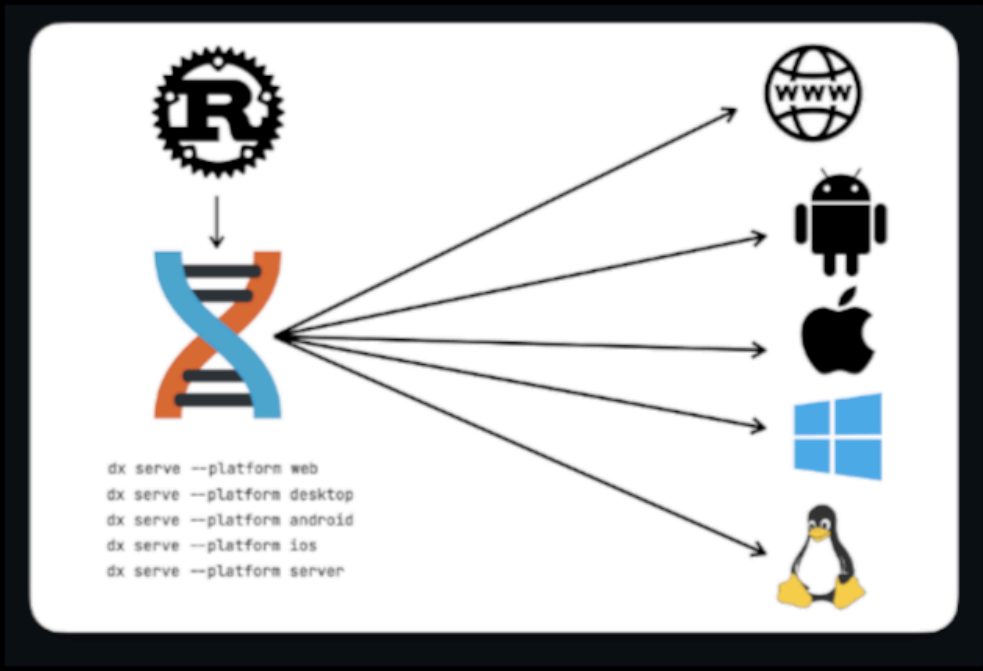
Why Dioxus?

We built Dioxus because we believe the
Developers need to learn and install doz
world. We need a simpler and more pow



What is Dioxus?

Dioxus is a developer-friendly framework that empowers developers to ship cross-platform apps with one codebase. You write your apps in Rust, style them with HTML/CSS, enhance them with native APIs, and distribute them as platform-native bundles.



In many ways, Dioxus is similar to Flutter: we integrate our own build tools, foster an ecosystem, and provide a GUI framework. In key areas, Dioxus takes a different approach:

- Apps are declared with HTML and CSS instead of a custom styling solution
- Reactivity is inspired by web frameworks like React and SolidJS
- Dioxus code runs natively with no virtual machine, enabling zero-overhead calls to system APIs

Docs Blog Casts

Build apps locally and save lots

No need for SaaS or cloud services. Save time, money, and effort . . . auth, workers, emails & more out of the box.

```
// Literals
format::text("Loco")

// Tera view engine
format::render().view(v, "home/hello.html", json!({}))

// strongly typed JSON responsed, backed by `serde`
format::json(Health { ok: true })

// Etags, cookies, and more
format::render().etag("loco-etag")?.empty()
```

Views

Use server-rendered templates with Tera or JSON. Loco can render views on the server or work with a frontend app seamlessly. Configure your fullstack set up any way you like.

```
impl Model {
 pub async fn find_by_email(db: &DatabaseConnection, email: &str)
 -> ModelResult<Self> {

    Users::find()
      .filter(eq(Column::Email, email))
      .one(db).await?
      .ok_or_else(|| ModelError::EntityNotFound)
 }

 pub async fn create_report(&self, ctx: &AppContext) -> Result<()> {
    ReportWorker::perform_later(
      &ctx,
      ReportArgs{ user_id: self.id }
    ).await?;
 }
}
```

Models

Model your business with rich entities and avoid writing SQL, backed by SeaORM. Build relations, validation and custom logic on your entities for the best maintainability.

```
pub async fn get_one(
   respond_to: RespondTo,
   Path(id): Path<i32>,
   State(ctx): State<AppContext>,
) -> Result<Response> {
   let item = Notes::find_by_id(id).one(&ctx.db).await?;
   match respond_to {
    RespondTo::Html => html_view(&item),
    _ => format::json(item),
   }
}

pub fn routes() -> Routes {
   Routes::new()
    .prefix("notes")
    .add("/{id}", get(get_one))
}
```

Controllers

Handle Web requests parameters, body, validation, and render a response that is content-aware. We use Axum for the best performance, simplicity and extensibility.

```
jobs:
 db_vacc
  run: "d
  shell: t
  schedu
  tags: ["

 send_bi
  run: "u
  schedu
  tags: ["
```

Simplif

10

Take your Rust web development to the next level

Cot is a powerful, type-safe, and fully featured Rust framework, delivering top-notch security and blazing speed. Cot empowers you to build production-ready web apps in record time — without compromising on performance or reliability.

Get Started → View on GitHub

Less crate hunting, more building

A taste of what everyday Cot code looks like — type-safe, declarative, and free of ceremony.

Type-safe ORM queries

Define a model once, then query it with the `query!` macro — fields and types are checked at compile time, not at runtime.

```
#[model]
struct Customer {
    #[model(primary_key)]
    id: Auto<i64>,
    #[model(unique)]
    email: Email,
    is_verified: bool,
}

let customer = query!(Customer, $email == email)
    .get(&db)
    .await?;
```

Forms that validate themselves

`#[derive(Form)]` turns a struct into a fully validated HTML form, complete with types like `Email` and `Password` that guarantee well-formed data.

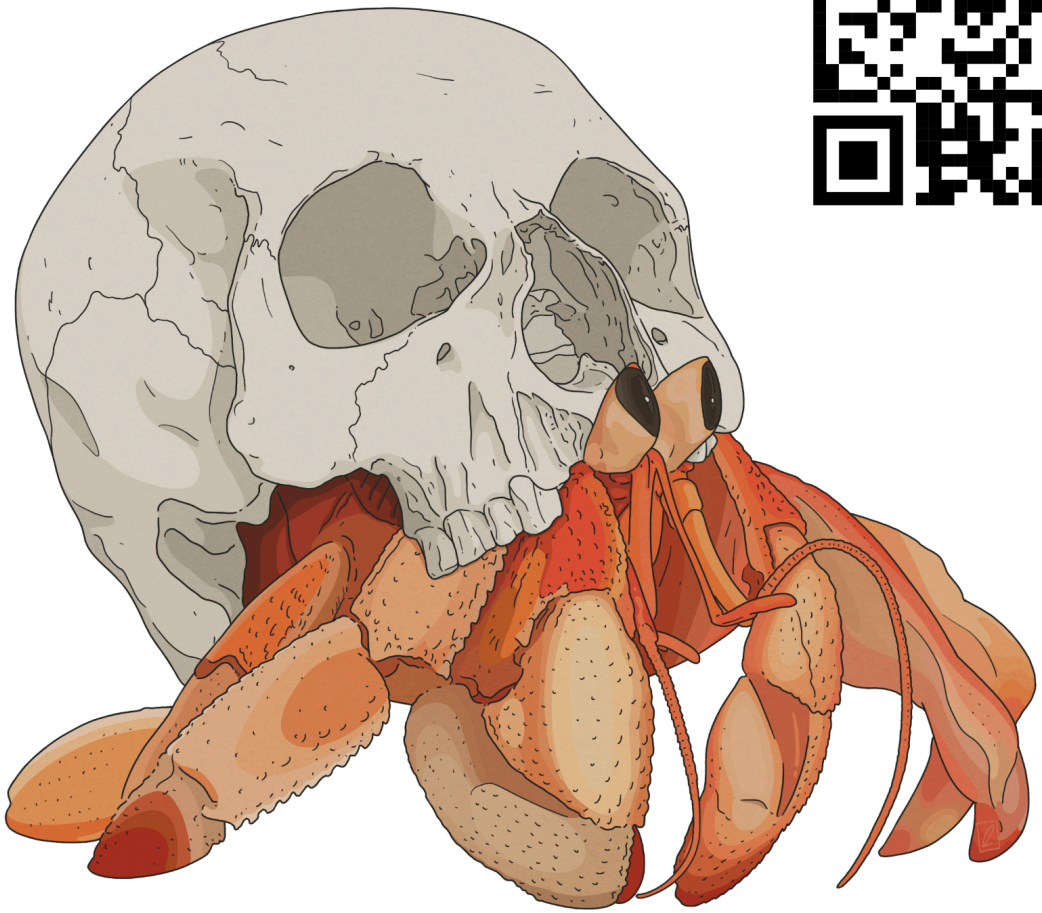
```
#[derive(Debug, Form)]
struct SignupForm {
    #[form(opts(max_length = 100))]
    username: String,
    email: Email,
    password: Password,
}

async fn signup(
    RequestForm(form): RequestForm<SignupForm>,
) -> cot::Result<Response> {
    let form = form.unwrap();
    // username, email, and password are already validated
    DatabaseUser::create_user(&db, &form.username, &form.password).await
    Ok(reverse_redirect!(urls, "index")?)
}
```

Si vous voulez aller plus loin...

(parce qu'il y en a encore plein, des dingueries comme ça 😅, dans Rust)

ZERO TO PRODUCTION IN RUST

AN OPINIONATED INTRODUCTION TO BACKEND DEVELOPMENT



Learn Rust With Entirely Too Many Linked Lists

I fairly frequently get asked how to implement a linked list in Rust. The answer honestly depends on what your requirements are, and it's obviously not super easy to answer the question on the spot. As such I've decided to write this book to comprehensively answer the question once and for all.

In this series I will teach you basic and advanced Rust programming entirely by having you implement 6 linked lists. In doing so, you should learn:

- The following pointer types: `&`, `&mut`, `Box`, `Rc`, `Arc`, `*const`, `*mut`, `NonNull` (?)
- Ownership, borrowing, inherited mutability, interior mutability, Copy
- All The Keywords: struct, enum, fn, pub, impl, use, ...
- Pattern matching, generics, destructors
- Testing, installing new toolchains, using `miri`
- Unsafe Rust: raw pointers, aliasing, stacked borrows, UnsafeCell, variance

Yes, linked lists are so truly awful that you deal with all of these concepts in making them real.

Everything's in the sidebar (may be collapsed on mobile), but for quick reference, here's what we're going to be making:

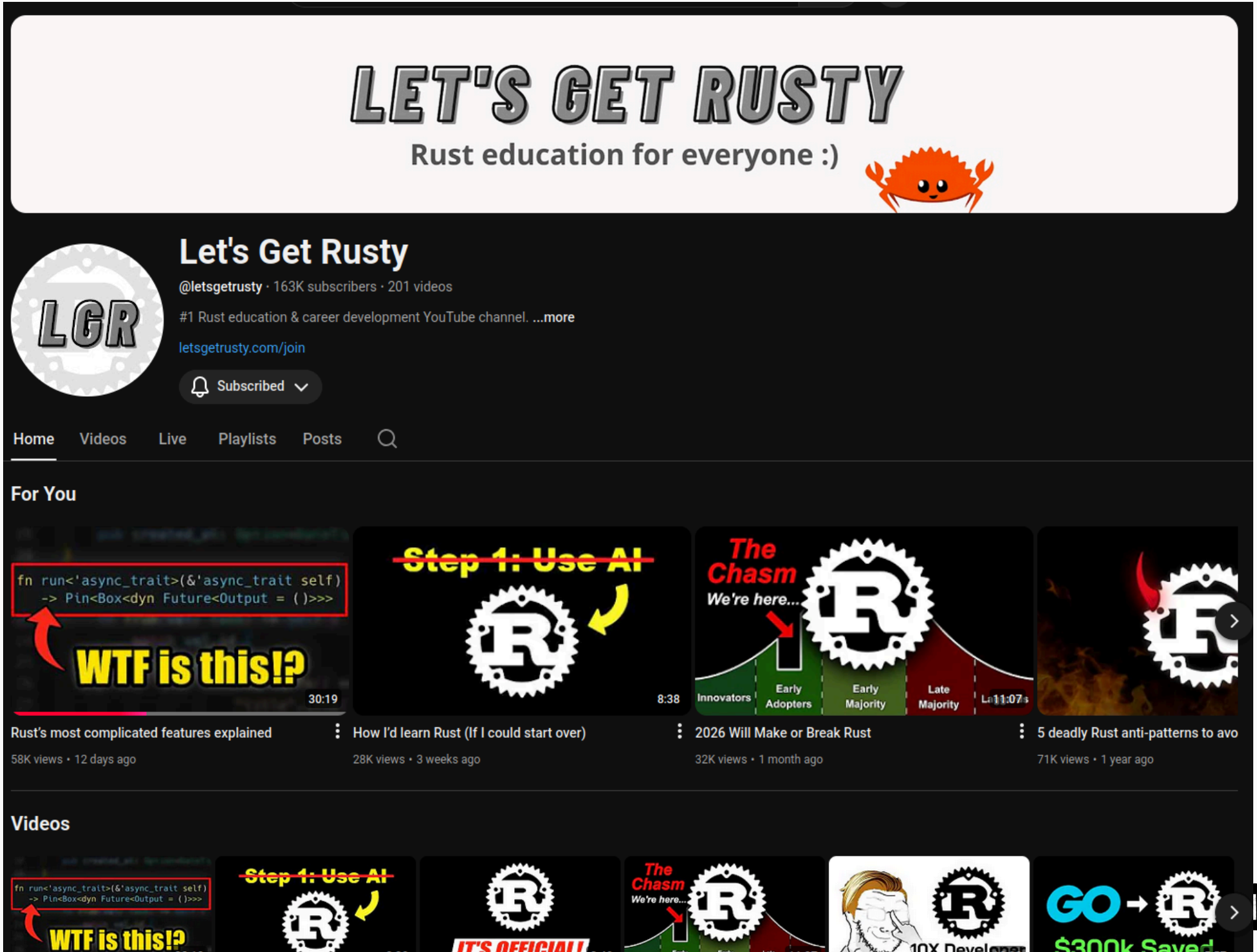
1. A Bad Singly-Linked Stack
2. An Ok Singly-Linked Stack
3. A Persistent Singly-Linked Stack
4. A Bad But Safe Doubly-Linked Deque
5. An Unsafe Singly-Linked Queue
6. TODO: An Ok Unsafe Doubly-Linked Deque

Créer une application hypermédia en Rust

Aperçu de l'application

L'application qui sert d'exemple dans Hypermedia Systems est une application CRU
qui gère une liste de contacts. Le livre montrer comment créer une application
hypermédia qui offre bonne interface utilisateur, en alternative aux applications
monopages. Grâce à une minuscule bibliothèque JavaScript (htmx) et à quelques
attributs déclaratifs sur les éléments HTML, toutes sortes d'interactions utilisateur
peuvent récupérer des fragments de code HTML depuis le serveur et ne modifier q
certaines parties de la page. Par exemple, quand on passe d'une page à une autre da

YouTube: Let's Get Rusty !



Stéphane TRÉBEL

Le Permacodeur

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Rust 🦀

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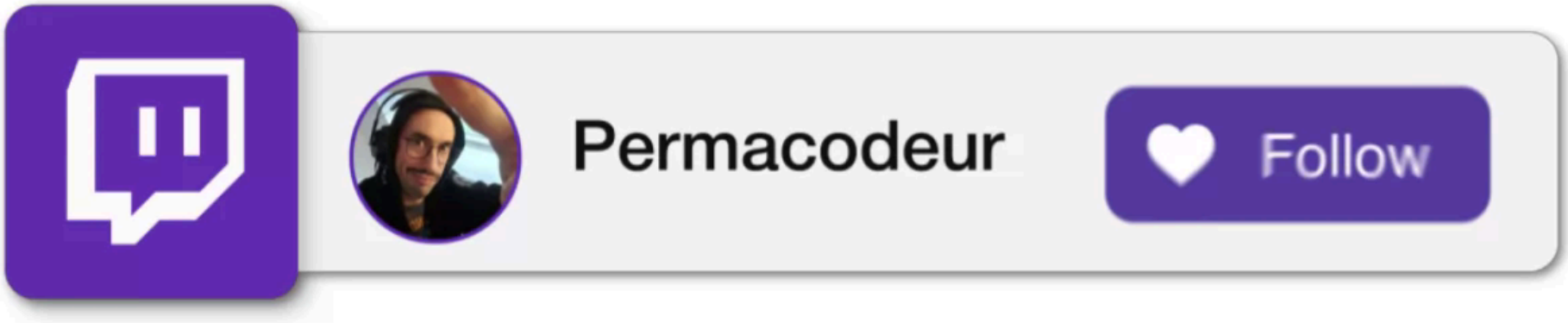
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🤩 Advent Of Code 2025...in Rust ! Jour 8/12 - Partie 1 🐙

```
fold(junction_box_list: &Vec<Coords>, connection_count: usize) -> usize {
    let mut nearest_list: BTreeMap<Coords, Coords> = BTreeMap::new();
    let mut nearest_list_sorted: HashMap<(Coords, Coords), i64> = HashMap::new();
    let max_distance: i64 = i64::MAX;

    for junction_box: &(u32, u32, u32) in junction_box_list {
        for other_box: &(u32, u32, u32) in junction_box_list {
            if junction_box < other_box {
                nearest_list_sorted.insert(
                    k: (*junction_box, *other_box),
                    v: compute_euclidian_distance(first_box: junction_box, second_box: other_box),
                );
            } else if junction_box > other_box {
                nearest_list_sorted.insert(
                    k: (*other_box, *junction_box),
                    v: compute_euclidian_distance(first_box: junction_box, second_box: other_box),
                );
            }
        }
    }
    let nearest_list_sorted = ne

    println!("Nearest list: {
    println!();
    println!("Nearest list le

    println!("Nearest list, s
    for (couple: &((u32, u32,
        println!("{distance}
    }
```

```
nearest_list_sorted HashMap<((u32, u32, u32), (u32, u32, u32)), i64, RandomState> v [LS]
nearest_list BTreeMap<(u32, u32, u32), (u32, u32, u32), Global>                    v [LS]
needs_drop()~(use std::intrinsics::needs_drop) const fn() -> bool                  f [LS]
needs_drop()~(use std::mem::needs_drop) const fn() -> bool                         f [LS]
net(use core::net)                                                                 M [LS]
net(use std::net)                                                                  M [LS]
net(use std::os::linux::net)                                                       M [LS]
net(use std::os::unix::net)                                                        M [LS]
new                                                                                [A]
```

```
INSERT  trunk!  src/main.rs[+]  rust-analyzer                rust  utf-8[unix]  36% ln:123/334≡%:33  W:2(L104)
-- INSERT --
```

```
   test result: FAILED. 0 passed; 1 failed; 0 ignored; 0 measured; 3 filtered out; finished in 0.00s

stderr ———

   thread 'tests_fold::fold_sample' (1550824) panicked at src/main.rs:310:9:
   assertion `left == right` failed
     left: 28
    right: 40
   note: run with `RUST_BACKTRACE=1` environment variable to display a backtrace

Cancelling due to test failure: 1 test still running
Cancelling [ 00:00:06] [                                        ]          3/4: 1 running, 0 passed, 0 skipped
           [ 00:00:06] aoc2025::bin/aoc2025 tests_fold::fold_gouz
```

Cabinet

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Mes liens 🤓

> *Rust 🦀 a désormais tout ce qu'il faut pour délivrer de la valeur pour vos applications métier. Jetez-y un œil ! 👀*